

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 3

The Map of Mathematical Fields

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. Why is it significant that you need continuous mathematics (analysis) to understand the distribution of prime numbers (discrete objects)?

2. Someone gives you a dataset and asks: “Are these two clusters actually different?” Name two mathematical fields that offer tools to answer this question, and explain what each field’s version of “different” means.

3. You are told that a certain property of a physical system is “topologically protected.” Without knowing the specific system, what can you infer about the property’s robustness? What would it take to destroy it?

4. A biologist says eigenvalues are “numbers that come out of matrices.” A physicist says eigenvalues are “natural frequencies of a system.” A geometer says eigenvalues are “scaling factors along principal axes.” Are they disagreeing?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
